Technical Progress Report

Project name:	**Biped robot**		
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1. Executive Summary

This report provides an update on the technical progress made in the past week for the Biped Robot project. Key developments include the completion of the overall structural scheme design of the biped robot, the determination of key component selection, the establishment of the project implementation timeline, and the preparation work for the subsequent mechanical modeling and hardware setup stages. The project is currently in the foundational preparatory phase, laying a solid physical and theoretical foundation for the realization of stable walking and subsequent intelligent upgrade of the biped robot.

2. Objectives & Scope

- Complete the mechanical structure design and hardware setup of a 6-degree-of-freedom biped robot, and achieve a stable straight-line walking posture to lay the physical foundation for the project.
- Based on the basic biped robot platform, conduct in-depth research on complex gait planning algorithms and environmental perception capabilities, and realize obstacle avoidance and specific object recognition functions.
- Establish a complete development and testing process for the biped robot, and optimize the control algorithm and mechanical structure to improve the stability and naturalness of the robot's walking.

3. Technical Work Completed

3.1 Overall Scheme Design

- Determined the **6-degree-of-freedom bipedal structure design** for the robot, with 3 core degrees of freedom for each leg (hip joint forward-backward swing, hip joint left-right swing, knee joint bending), forming a minimalist and practical mechanical architecture.
- Analyzed the advantages of the selected structural scheme: while ensuring basic movement capabilities, it significantly reduces the complexity of mechanical design and control algorithms, which is suitable for the project's research and development orientation.

3.2 Key Component Selection & Confirmation

- Completed the screening and determination of all core components of the biped robot, covering control & vision, sensor & drive, power joint, and energy & power supply categories, with clear models and specifications for each component:

- **Control and Vision**: ESP32 DevKit V1 development board (core control), ESP32-CAM (with OV2640 camera, visual module)
- **Sensor and Drive**: GY-521 MPU6050 posture balance sensor, PCA9685 16-path 12-bit multi-channel servo driver board
- **Power Joints**: 6 MG996R medium-sized main steering gears, 4 MG90S miniature auxiliary servo motors
- **Energy and Power Supply**: LM2596S high-power voltage reduction module (with digital display), LB218B 7.4V lithium battery pack (main power supply)

- Verified the matching degree between each component, ensuring the rationality of component collocation for subsequent hardware assembly and functional realization.

3.3 Project Planning & Task Decomposition

- Established a phased implementation plan for the project, dividing the follow-up work into **Mechanical Modeling, Hardware Setup, Algorithm Development, Integration Testing, and Optimization Acceptance** stages, and clarified the core objectives of each stage.
- Completed the detailed task breakdown and time planning for the key early stages (Mechanical Modeling and Hardware Setup), and formulated a 2-week overall development timeline with clear division of core work for each time period.
- Carried out comprehensive data collection and analysis on the biped robot's mechanical structure, electronic control system, and software algorithms, laying a complete theoretical foundation for subsequent practical development.

4. Observations & Findings

4.1 Scheme Advantages & Feasibility

- The selected 6-degree-of-freedom minimalist structure has high practicality for the project's initial goal of "stable straight-line walking", and the low design complexity can effectively reduce the difficulty of subsequent mechanical modeling and control algorithm development, with strong implementability.
- The selected key components have the characteristics of high cost performance, mature technology, and easy matching. The specifications of each component (size, power, interface) are compatible, which can ensure the smooth progress of hardware assembly and system debugging.

4.2 Project Progress & Preparatory Status

- The project has completed all foundational preparatory work in the first week, including scheme design, component selection, and timeline formulation, and the progress is consistent with the expected plan.
- The theoretical research on mechanical structure and control algorithms has been completed, and the technical route for subsequent mechanical modeling (based on SolidWorks) and hardware development is clear, with no obvious technical conflicts or matching problems found in the early stage.

4.3 Stage Goal Clarity

- The phased task decomposition makes the project goals at each stage clear and measurable: the short-term goal of "foundation and walking" and the long-term goal of "intelligent upgrade" are closely connected, and the work of each stage serves the overall project objectives, forming a complete development logic.

5. Challenges & Roadblocks

- Lack of practical experience in 3D mechanical modeling and virtual assembly of biped robots; it is necessary to master the operation of SolidWorks software in a short time to ensure the rationality of component modeling and avoid spatial interference during assembly.
- The initial gait planning algorithm is only in the theoretical design stage; the transformation from theoretical algorithm to actual code and the adaptation with hardware have unknown difficulties, and the real-time response and control accuracy of the system need to be verified.
- The subsequent integration of environmental perception modules (ultrasonic, infrared, visual recognition) with the basic walking platform has potential technical difficulties.
- The coordination between perception and motion control needs to be further explored.

6. Next Steps & Recommendations

- Conduct SolidWorks software foundation learning, master the core operation logic and basic commands of the modeling software to meet the needs of robot component modeling.
- Complete the 3D model design of all key components of the robot (such as lower legs, hip joints) based on the determined structural scheme, and carry out virtual assembly of components to check and optimize spatial interference problems.

7. Conclusion

- In the first week of the project, significant foundational progress has been made in the biped robot research and development, including the completion of the 6-degree-of-freedom structural scheme design, the determination of all key core components, the formulation of a detailed phased implementation plan and 2-week development timeline, and the completion of comprehensive theoretical research on mechanical structure and control algorithms.
- The project is currently in a good preparatory state, and all early work has laid a solid physical and theoretical foundation for the subsequent mechanical modeling, hardware setup and stable walking realization of the biped robot.